

Nexus Document Platform 4.2 — Deployment and Scaling

Deployment

The platform is deployed as containers. API Gateway, Document Processing Service, and Metadata Service are independently scalable. PostgreSQL and object storage are managed infrastructure components.

Scaling

API Gateway scales horizontally based on request rate. Metadata Service scales based on queue depth and processing latency. Document Processing Service scales based on CPU and active jobs.

Health Checks

Each service exposes `/health/live` and `/health/ready`. Liveness indicates the process is running. Readiness verifies required dependencies.

Failure Handling

Message consumers use exponential backoff. After five unsuccessful processing attempts, a message is moved to the dead-letter queue. Operators can replay dead-letter messages after correcting the underlying problem.

Related Components

See `deployment.yaml`, `retry-policy.md`, and `troubleshooting.txt`.